



HMDKI-640/27.6-□

REACTANCE ON-LOAD TAP CHANGER OPERATING INSTRUCTION

SHANGHAI HUAMING POWER EQUIPMENT CO.,LTD.

Thank you for using our company's HMDKI-640/27.6-□ reactance type on-load tap-changer.

Please make sure you will read this “operating instruction” carefully before operating the on-load tap-changer.

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1. General

HMDKI-640/27.6-□ Reactance Type On-Load Tap-Changer (Hereinafter referred to as OLTC) is for application of on-load voltage regulating of the single phase step voltage regulator.

HMDKI-640/27.6-□, a single-phase OLTC, features as an external structure and is installed inside the external oil tank of transformer by a flange fitted on the aluminum base of the end of OLTC. This type of OLTC has optionally three forms of voltage regulation in accordance with different type of connections with transformers, such as without change-over selector, or with reversing switch, or with coarse & fine tap selector.

The outline of OLTC is shown in fig.1 and 2, the principle of voltage regulation in fig.3: operating position (at set position) table in fig. 4 and 5, and basic connection diagram in fig. 6. This operating instruction contains the specification on how to use and install the OLTC.

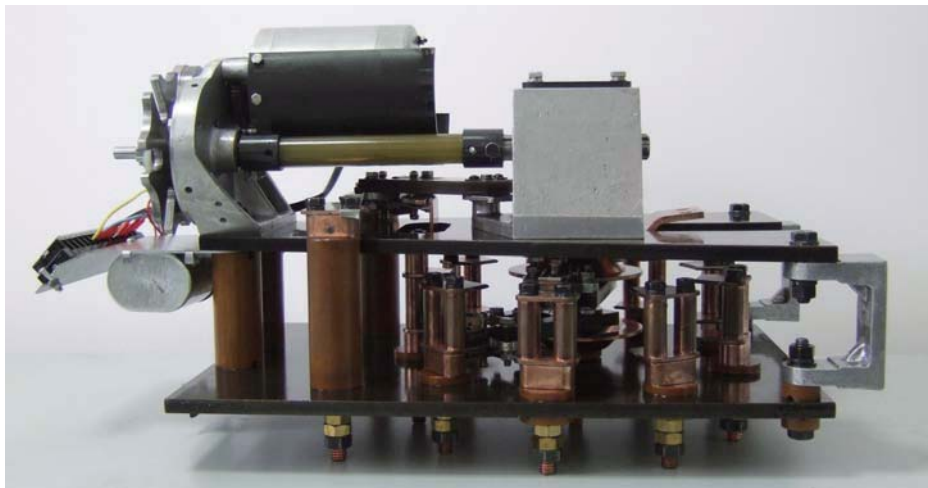


Fig. 1 Outline view of OLTC

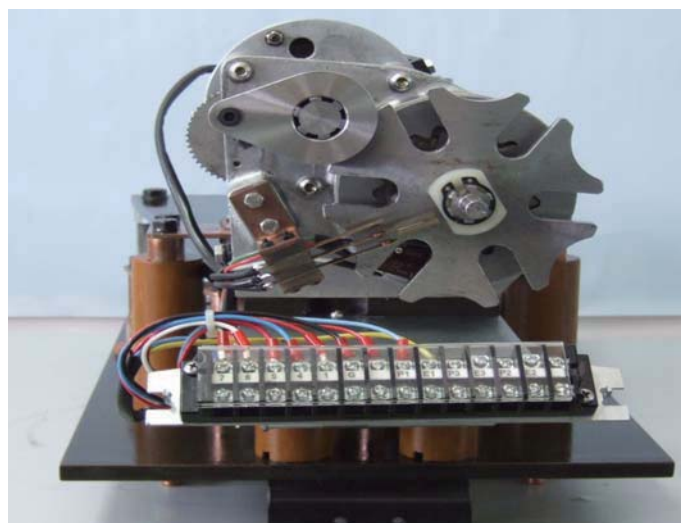
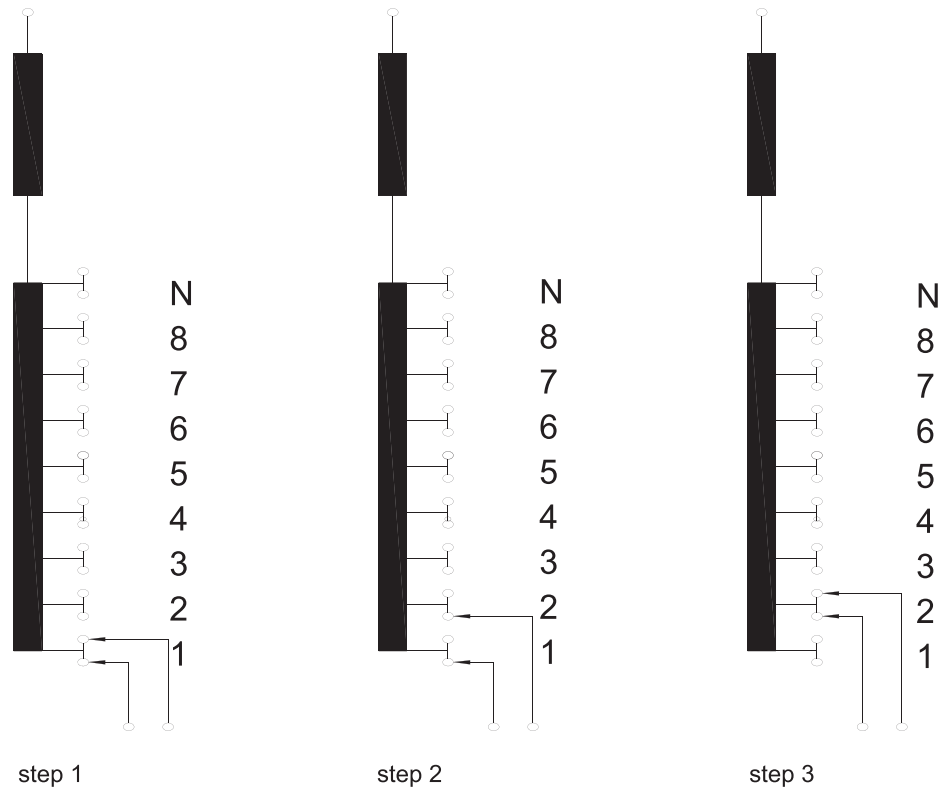


Fig. 2 Outline view of OLTC



1. Without change-over selector

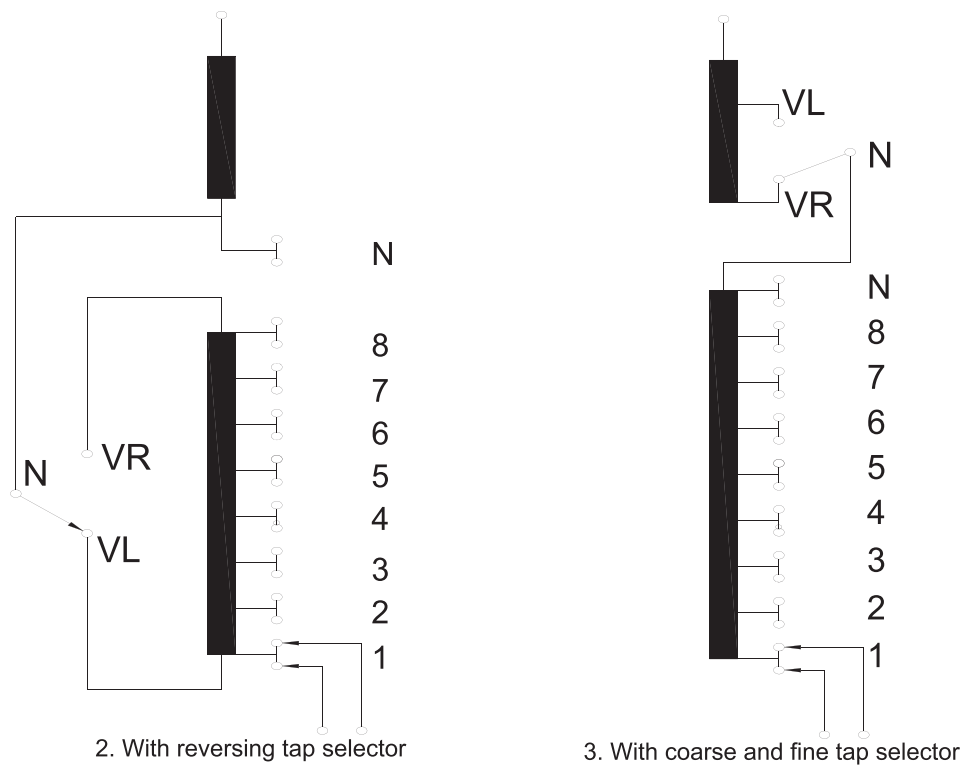
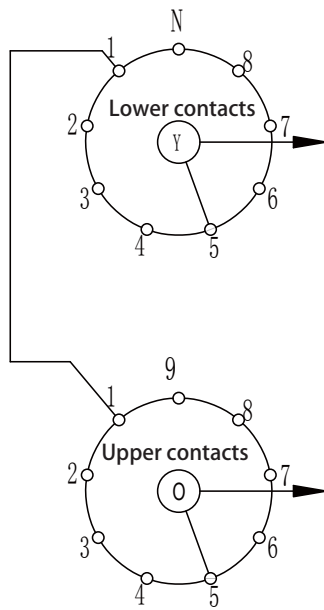


Fig.3 The principles of voltage regulation



Steps	16
Posistions	17
Set position	°

Tap selector contact position		Display position
Upper	Lower	
9	9	17
9	8	16
8	8	15
8	7	14
7	7	13
7	6	12
6	6	11
6	5	10
5	5	9
5	4	8
4	4	7
4	3	6
3	3	5
3	2	4
2	2	3
2	1	2
1	1	1

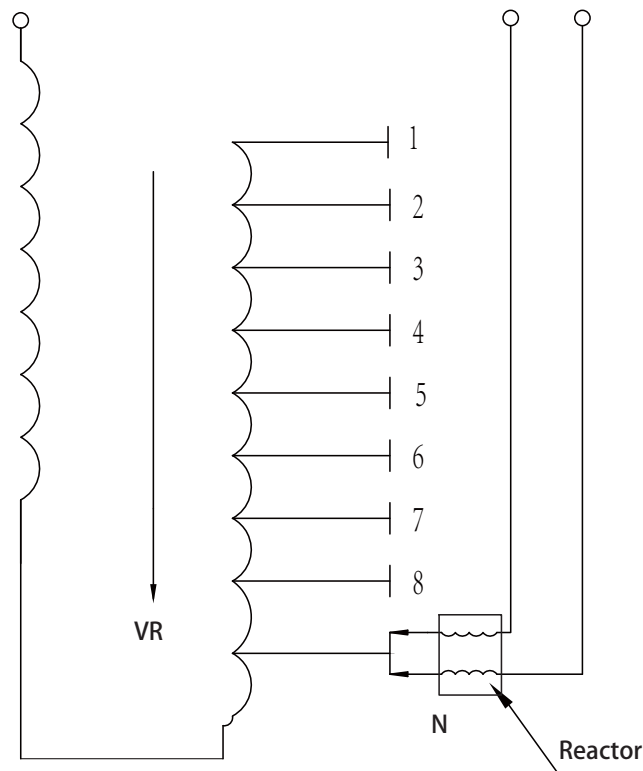


Fig. 4 Operating position table and connection diagram without change-over selector

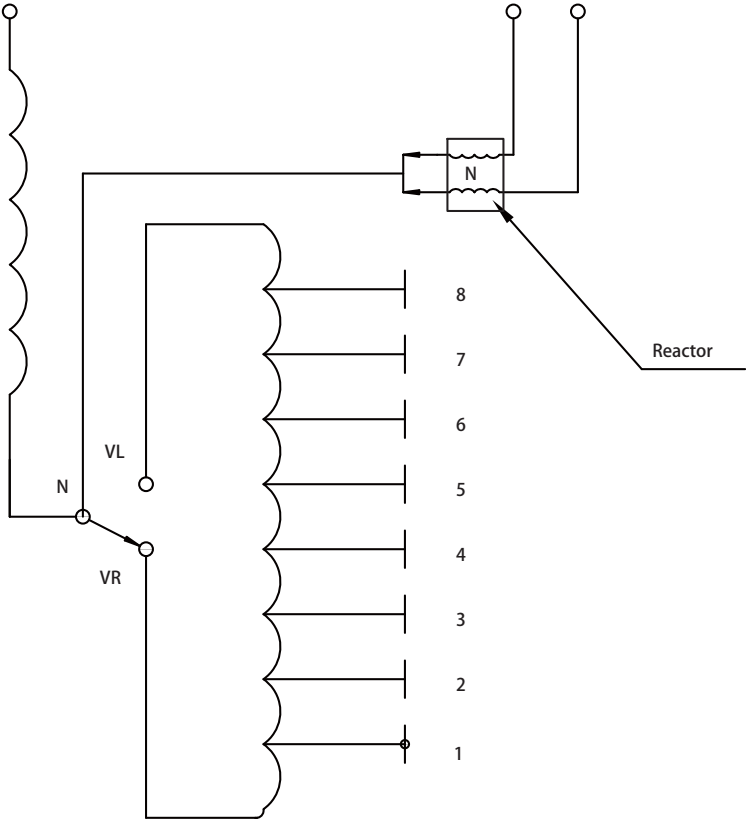
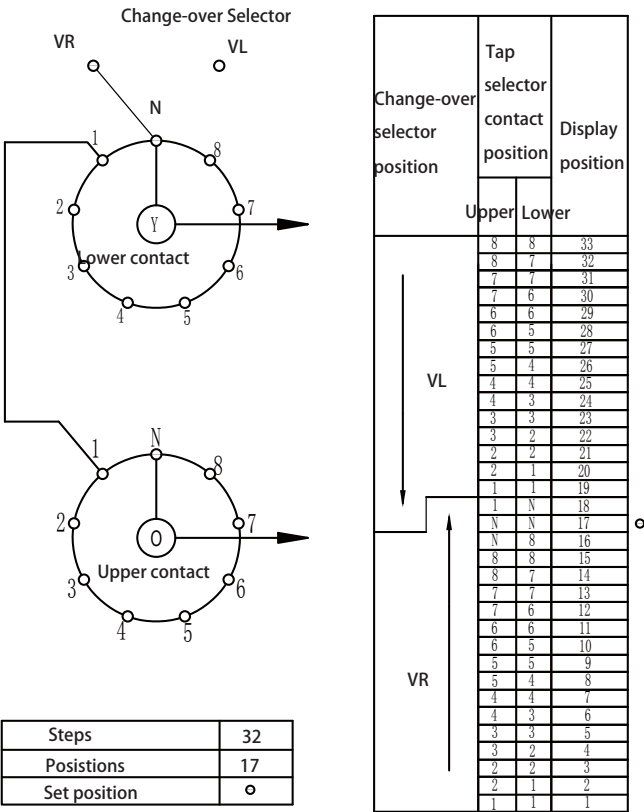
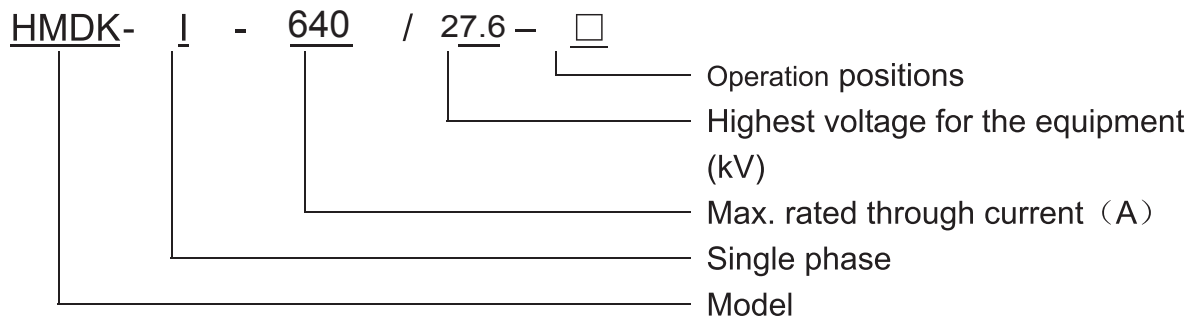


Fig. 5 Operating position table and connection diagram with change-over selector



Fig. 6 Basic connection diagram

1.1 Model designation:



1.2 Usage and scope of application

This type of OLTC is suitable for the single-phase voltage regulator which the maximum rated current is 668A and the highest voltage for equipment 27.6kV, frequency 50-60Hz. The maximum number of service position is 17 without change-over selector or 33 with change-over selector.

1.3 Operating conditions

- 1.3.1 The operation temperature of OLTC in the oil should be not higher than 90°C and not lower than -30°C.
- 1.3.2 If the OLTC is stored open in the air, the ambient temperature should be not higher than 40°C and not lower than -30°C.
- 1.3.3 Vertical inclination of OLTC installed in the transformer must not be higher than 2%.
- 1.3.4 Any explosive, corrosive gases or dust must not be present at site of the OLTC.

2. Main technical data

Table 1 Main technical data

Rated current (A)			640
Max. rated through current (A)			668
Rated short-circuit withstand current (KA)		2S _{rsm}	16
		0.8S _{rsm}	20
		Peak value	45.2
Rated voltage (KV)	Max.		27.6
	Min.		2.5
Step Voltage (V)			Rated voltage*0.625%
Insulation to ground	Rated lightning impulse withstand voltage 1.2/50 μ s	Below 13.8KV	95KV
		Above 13.8KV	200KV
	Rated power frequency withstand voltage 50Hz, 1min		70KV
Number of Phases			Single phase
Frenquency (Hz)			50 or 60
Max. Operating positions			33
Mechanical life			Not less than 2,000,000 times
Electrical life			Not less than 180,000 times
Operating temperature			-30℃～90℃
Drying temperature			125℃/16h
Time interval between operations			Not less than 1s
Weight (kg)			32
Motor Parameter	Rated voltage (V)		115
	Power (W)		58.7
	Frequency (Hz)		60
	Current (A)		0.46

3. Structure

The OLTC has an external structure and consists of damped motor, driving system, control system, movable contacts, stationary contacts, upper and lower insulating panels, aluminum bracket, etc.

- 3.1 The damping parts of damped motor are composed of stationary friction plate, movable friction plate, compressed spring, and screening shield, etc.
- 3.2 The driving system consists of gear unit, geneva wheel, coupling shaft, splined shaft, cam, upper and lower wheel disc, inner shaft and outer shaft, etc.
- 3.3 The control system includes inner and outer gear units, eccentric bushing, positioning parts, movable roller, idler pulley, elastic-stripped electrical contact, micro-switch, etc.
- 3.4 The movable contact is a clamping piece. The switching contact part is made of copper-tungsten alloy and the connecting contact is made of pure copper with silver plating. The bracket and back panel are made of aluminum alloy and laminated insulating board respectively.
- 3.5 The stationary contacts and the main contacts are made of pure copper with copper-tungsten alloy, and the slip ring is made of pure copper.
- 3.6 The parts and aluminum bracket mentioned above are installed on the upper and lower insulating panels, there is a bolting hole to connect the oil tank of the voltage regulator on the aluminum bracket. For details see figure 1 and 2.

4. Routine test program

- 4.1 The OLTC should be free of any mechanical failure after 1,000 times operation for mechanical test.
- 4.2 The contact pressure and contact resistance should be measured after final assembly, and the contact pressure should be $62 \pm 5\text{N}$, every pair of contacts should sustain same force, and the contact resistance should be $\leq 60\mu\Omega$.
- 4.3 The switching time of movable contacts should be range from 25 to 33ms.
- 4.4 Energized the tap changer with 1.2 time of Max. rated through current in transformer oil, the temperature rise should be no more than 20K.
- 4.5 The power frequency withstand voltage to ground should be 70 kV/ 1min under

frequency of 50Hz, and the impulse withstand voltage to ground should be 125kV (1.2/50 μ s).

- 4.6 The withstand short-circuit: thermal 16 (kA, 2s), dynamic 45.2 (kA, peak value)
- 4.7 The mechanical life should be more than 2,000,000 operations and electrical life more than 180,000 operations.

5. Storage and transportation

- 5.1 The warehouse for storing the OLTC should be dry and clean, in which the measures of preventing erosive gas and moisture should be taken, and the ambient temperature should not be higher than +40°C or lower than -30°C, and also the relative humidity should not exceed 85%.
- 5.2 The packing should be strong enough to protect the OLTC from dust and moisture and should be fixed firmly and handled carefully during the transportation.

6. Documents upon delivery

- 6.1 Qualified certificate.
- 6.2 Packing list.
- 6.3 Operating instructions.
- 6.4 Testing report

7. Accessories of the OLTC

- 7.1 This OLTC should be equipped with flexible shaft and position indicator, flexible shaft and position indicator should be supplied by Huaming. Please refer to appendix 2 for connection of OLTC and position indicator.
- 7.2 This OLTC is with the terminals connected to the control unit, please refer to appendix 3 of connection diagram between OLTC and the control unit, the control unit should be prepared by user.

8. Installation method

8.1 Inspection before installation

- 8.1.1 Check whether the model of OLTC meets the design requirements of voltage regulator and whether the certificate, operating instructions, packing list and other technical documents are complete.
- 8.1.2 Inspect whether any parts of the OLTC is defective or deformed.
- 8.1.3 Perform 10 operating cycles before installation to check whether there is any blocking for movement and damages for parts, and whether the service position is same as the position displayed on the indicator.
- 8.1.4 Measure contact resistance on each contact and compare with the value in the nameplate.

8.2 Installation

This tap changer, without oil chamber, can be placed inside the oil tank of voltage regulator. There are different installation locations depending on different tank structure.

Procedure of installation:

Clean all sealing surfaces before installation, and place an oil gasket on the installation flange of voltage regulator, and then lift the tap changer slowly into the tank of voltage regulator. Any damages against terminals and screen cover should be prevented during the lifting, and then confirm the installation position, finally, fix the OLTC on the flange of voltage regulator by 2-M12 screws.

Notes:

- a) Ensure all leads are reliable and tight.
- b) Ensure that there is no stress on the connecting leads after being attached to the OLTC.
- c) Connect the grounding wire between the head flange of OLTC and the cover of voltage regulator.
- d) Finally, a connection test must be performed before energizing the voltage regulator.

9. Drying process

Ensure the dielectric strength of OLTC by drying the OLTC with transformer, same process as the voltage regulator.

Notes:

- a) After drying, the OLTC is not permitted to be operated without oil. If necessary; all contacting points should be lubricated with transformer oil.
- b) After drying, the OLTC should be promptly immersed into transformer oil.
- c) Loose the ventilated screw on the head flange of OLTC as the voltage regulator

needs to fill the oil so that it could release the gas out.

- d) The newly-filled transformer oil must meet the requirements of dielectric strength ≥ 40 kV and water content ≤ 30 $\mu\text{L/L}$. The dielectric strength should be ≥ 30 kV and water content < 40 $\mu\text{L/L}$ when the OLTC is under operation.
- e) The OLTC is not permitted to be operated when the transformer oil can not meet the requirements. If in such case, the oil should be replaced or filtered.

10. Periodic maintenance

- 10.1 At least 3 cycles of de-energized operation should be performed every other year.
- 10.2 If the OLTC has kept running in one position for a long time, in case the OLTC is needed to adjust the position, several cycles of operations should be done before switching to the required position.
- 10.3 If re-connect the terminals of OLTC to taps of transformer windings, a serious inspection for connections must be carried out before putting the OLTC into use.
- 10.4 Check the connection of grounding wire.
- 10.5 Inspect and maintain the OLTC as required.

11. Overhaul and maintenance

An overhaul and maintenance must be carried out when the OLTC has run for a period of 5 years or operations of 50,000, which ever comes first.

12. Appendix

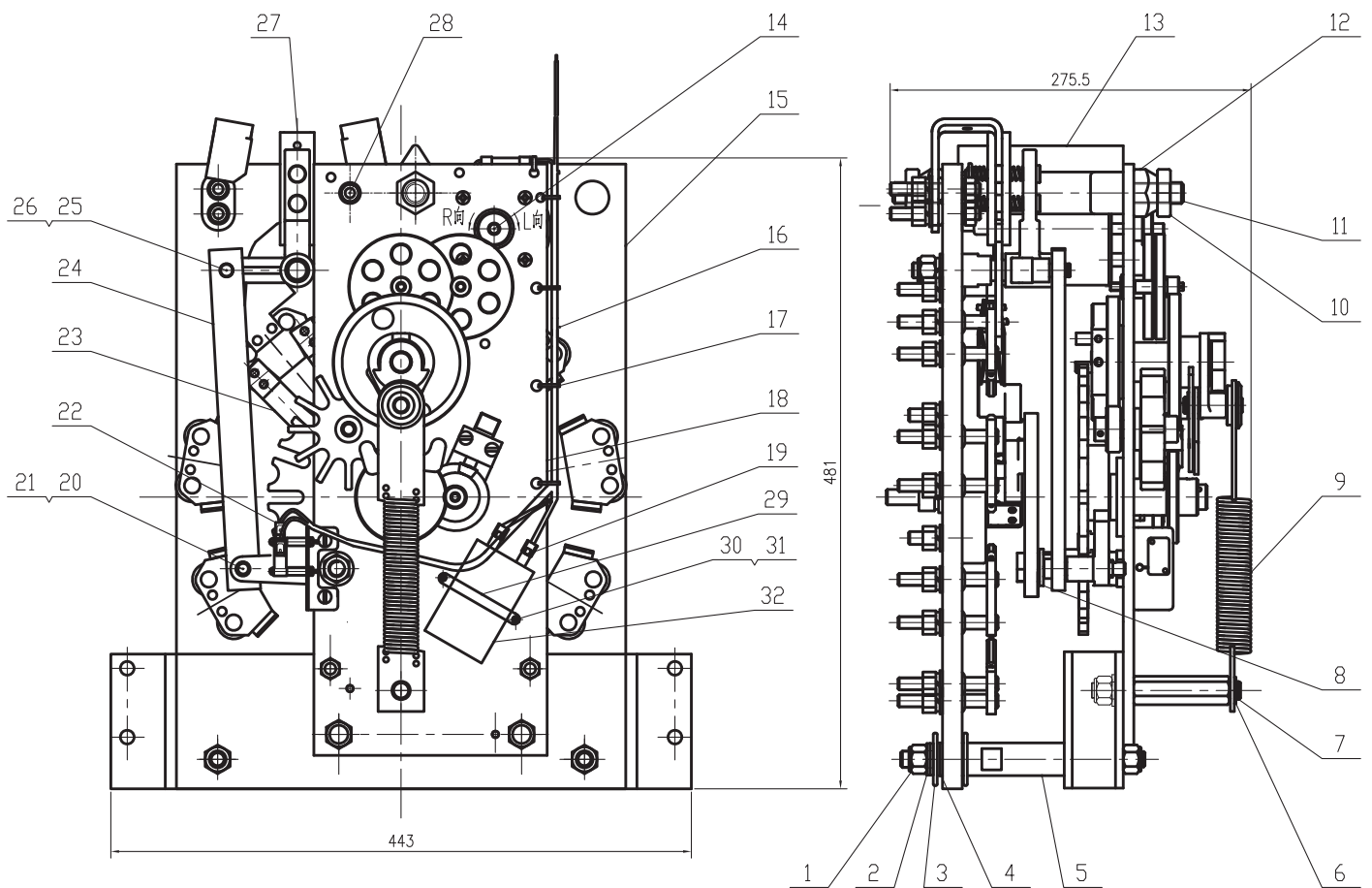
Appendix 1 Overall dimensions and installation schematics

Appendix 2 Electrical connection diagram of LTC and position indicator

Appendix 2 Electrical connection diagram of position indicator

Appendix 4 Connection diagram and setting position without change-over selector

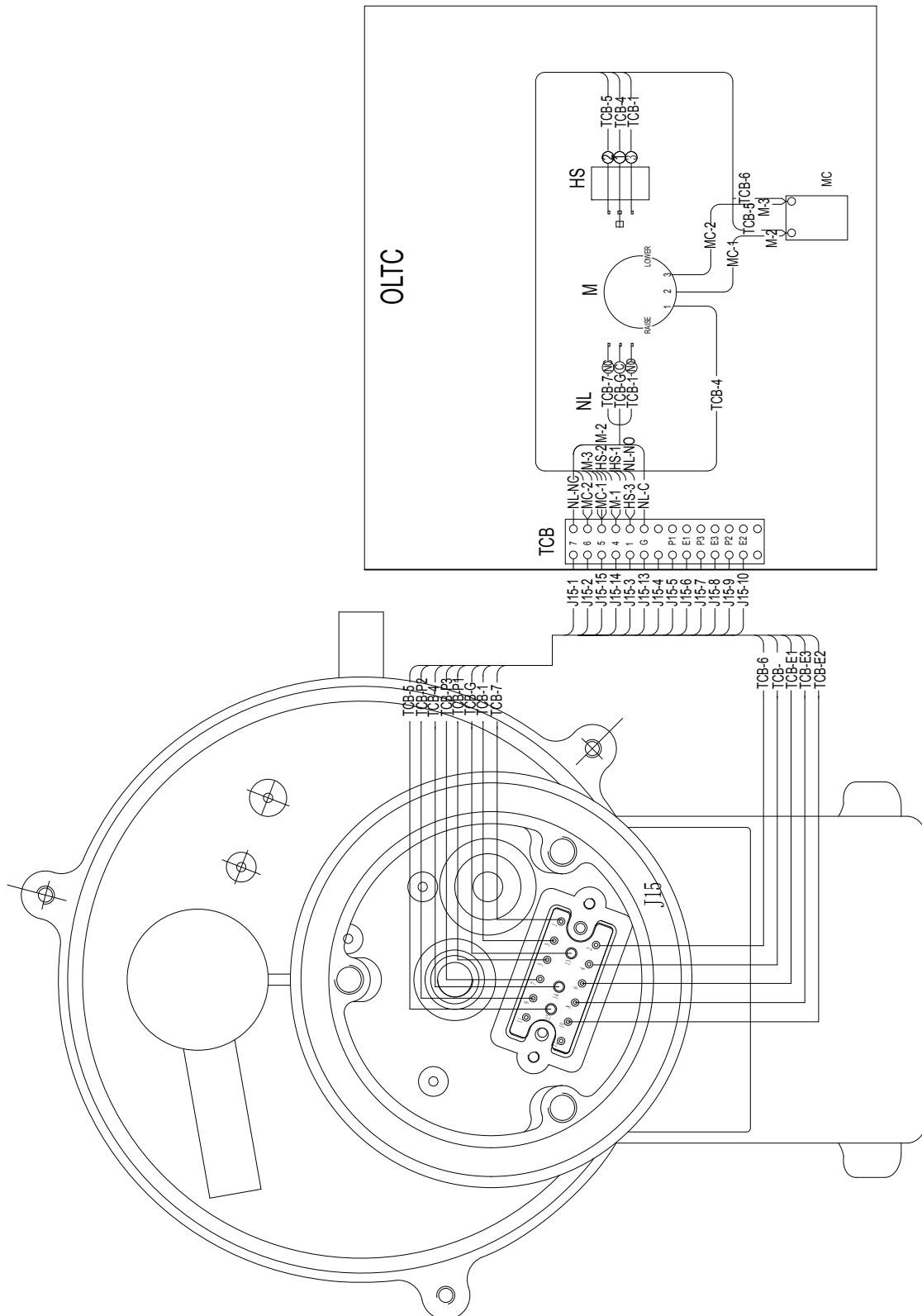
Appendix 5 Connection diagram and setting position with change-over selector



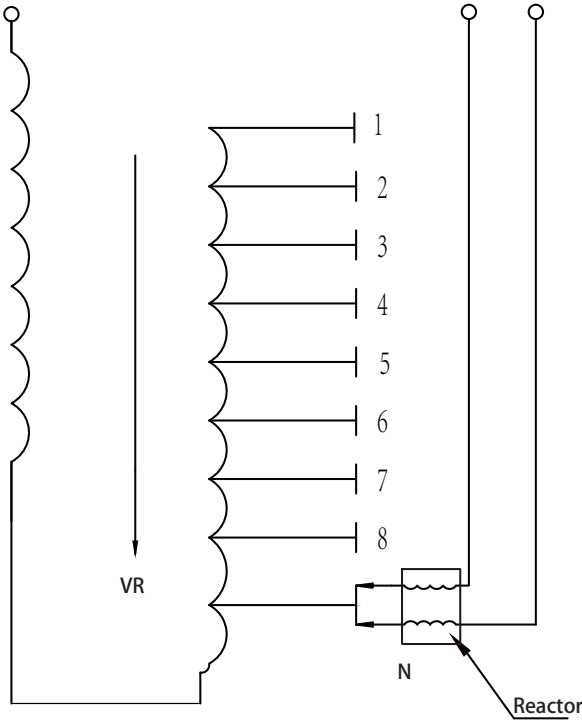
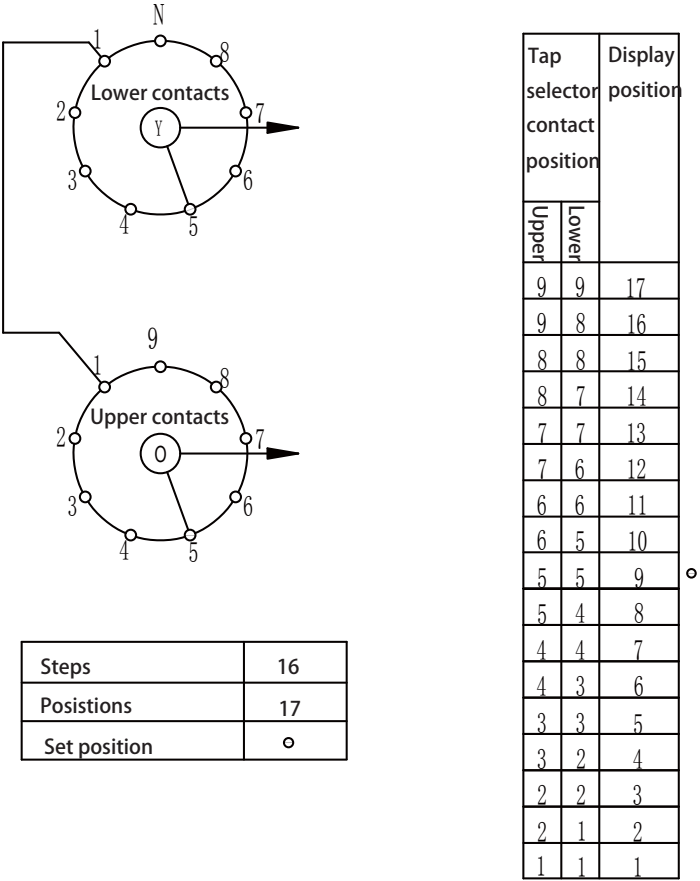
Overall dimensions and installation diagram

- | | | |
|----------------------------|-----------------------------------|-----------------------------------|
| 1. locknut 1/2-13UN | 13. Bracket | 25. Flat washer 10 |
| 2. disc spring | 14. Motor | 26. Shaft retaining ring 12 |
| 3. Washer 40 | 15. Stationary contact system | 27. Change-over contact |
| 4. Gasket | 16. Nylon binding tape | 28. Bolt M10x25UN |
| 5. Rod | 17. Wiring harness | 29. Clamp |
| 6. Washer 20 | 18. Driving plate | 30. Cross recessed pan head screw |
| 7. Shaft retaining ring 12 | 19. Straight insulation plug bush | 31. Locknut M5 |
| 8. Shaft sleeve | 20. Shaft pin | 32. Capacitor 8 μ F |
| 9. Spring | 21. Split pin 3x25 | |
| 10. Insulation nut | 22. Straight insulation plug bush | |
| 11. Stud | 23. Moving contact system | |
| 12. Insulation nut | 24. Pull rod | |

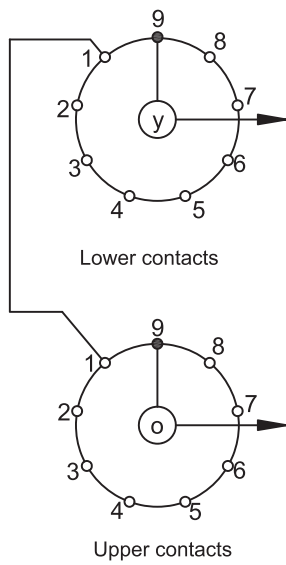
Appendix 1 Overall dimensions and installation schematics



Appendix 2 Electrical connection diagram of LTC and position indicator



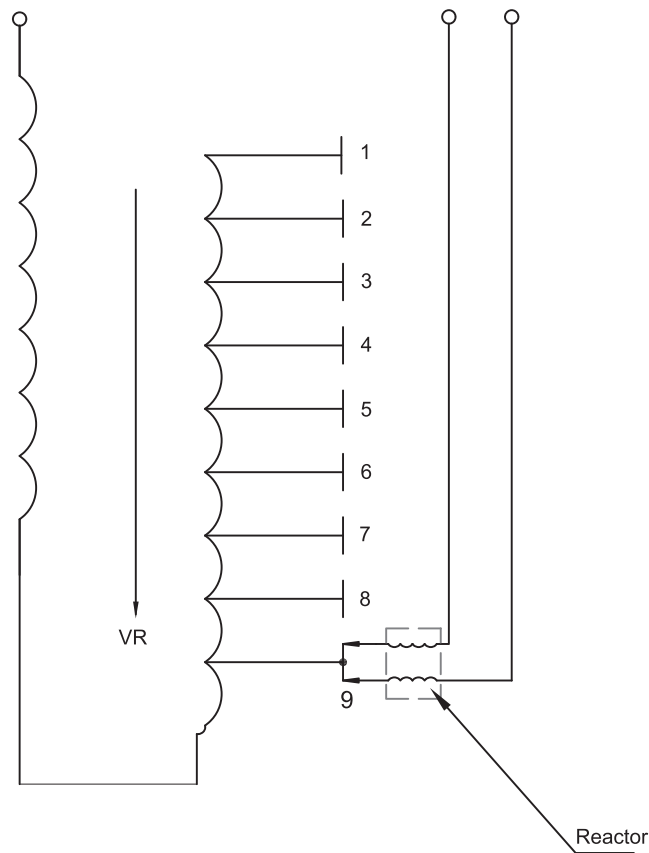
Appendix .3 Operating position table and connection diagram without change-over selector



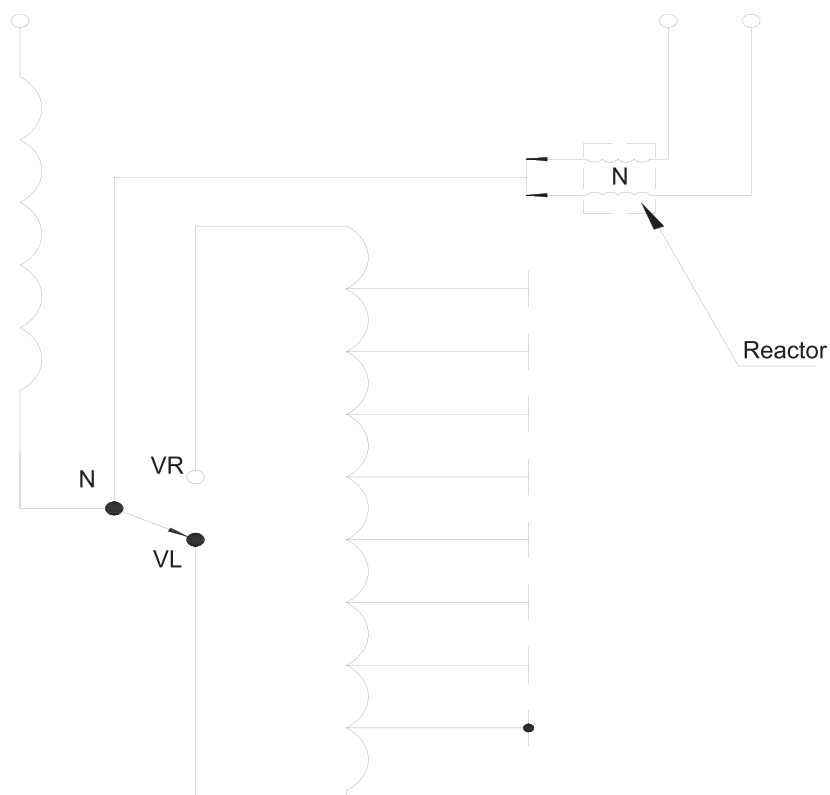
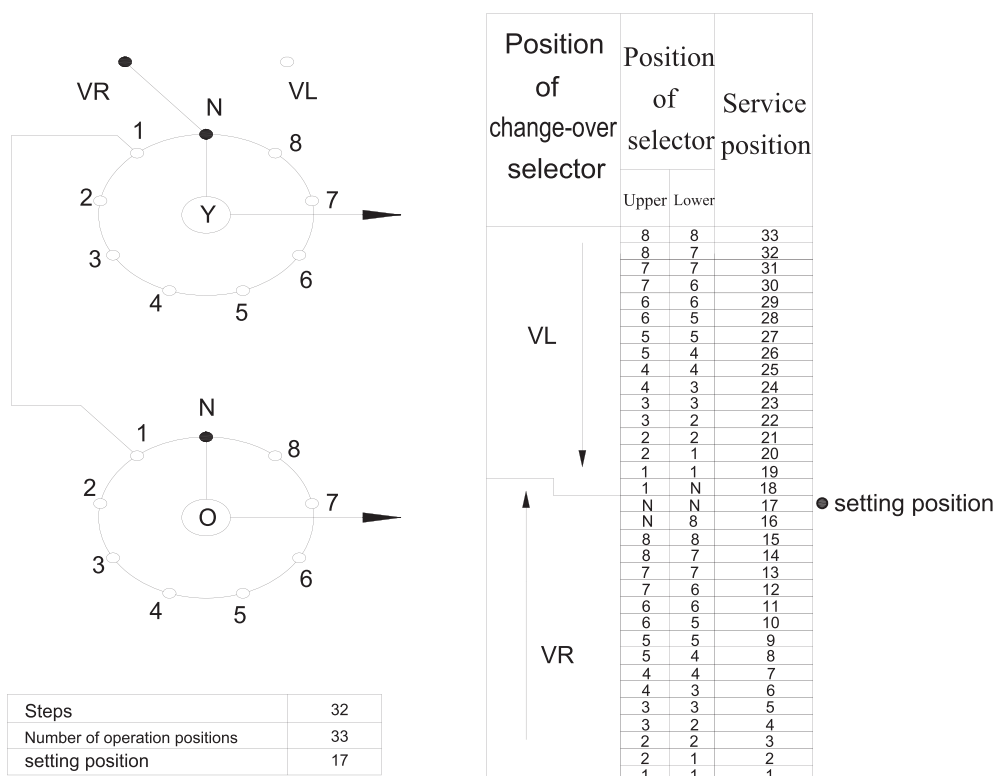
Steps	16
Number of operation positions	17
Setting position	9

Position of change-over selector		Service position
upper	lower	
N	N	17
N	8	16
8	8	15
8	7	14
7	7	13
7	6	12
6	6	11
6	5	10
5	5	9
5	4	8
4	4	7
4	3	6
3	3	5
3	2	4
2	2	3
2	1	2
1	1	1

● setting position



Appendix 4 Connection diagram and setting position without change-over selector



Appendix 5 Connection diagram and setting position with change-over selector

Notice of application and order

Customer and end user should keep records of the operation and inspection of the OLTC. In case of any special condition, please contact us and give us your report for data collecting purposes so that we can possibly give you further recommendations towards your next inspection.

The warranty period of the OLTC is 18 months from the date of shipment.

**We warmly welcome you to give us comment and suggestion for our products.
Thank you for your support and cooperation.**

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